Toward Robust Machine Learning based Network Intrusion Detection Systems

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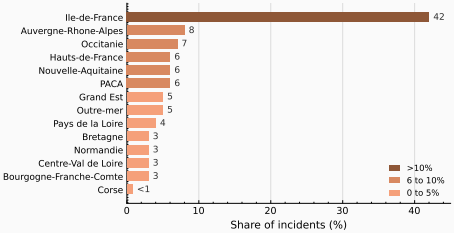
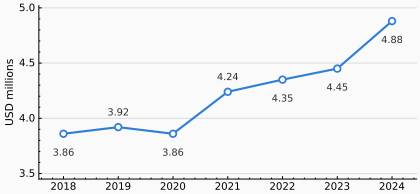
Introduction

Why this topic matters

- Modern societies increasingly rely on computer networks $\longrightarrow$ Higher exposure to cyber incidents.
- ANSSI 2024: **4,386** events handled (**+15%** vs. 2023); **1,361** confirmed incidents ($\approx$ **4/day**)
- Average data breach cost in 2024: **$4.88M**.
- In the French health sector, **35%** of reported incidents affected patient care.



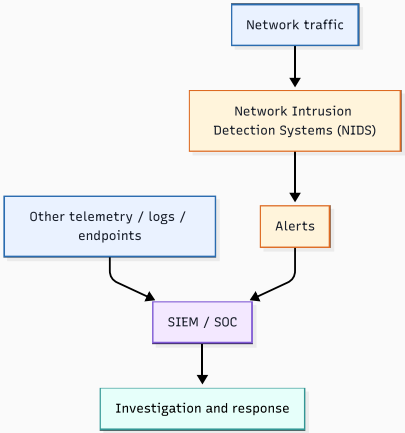
Source: **ANSSI, Panorama de la cybermenace 2024**



Source: **IBM Security, Cost of a Data Breach Report 2024**

Where NIDS fits in cyber defense

- Continuous monitoring to manage cyber risk
- SOCs and SIEM platforms centralize this monitoring
- Network traffic gives visibility into system communications
- NIDS turn traffic into alerts for investigation and response



⇒ **NIDS is one analytical layer inside the broader monitoring and response chain.**

From signatures to ML-based NIDS

Rule-based NIDS

- Tools: Snort, Suricata, Zeek
- Detect only known attack patterns or protocol violations, leaving unknown ones undetected.
- Encrypted traffic limits payload-based rules

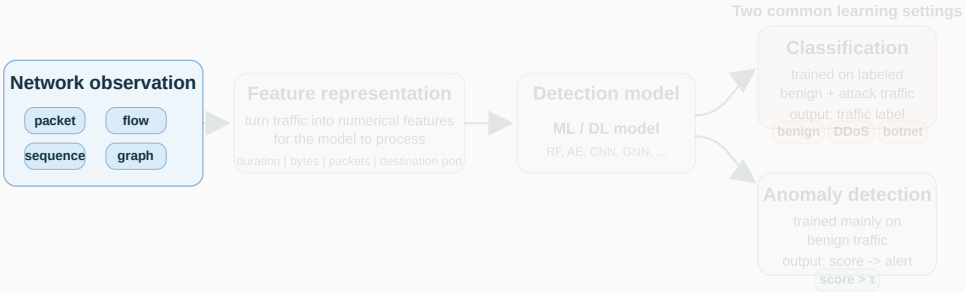
ML-based NIDS

- Example: Random Forest (RF) on flows
- Learn traffic patterns from historical data to generalize beyond exact rules.
- Can learn complex traffic patterns even under encryption

Operational reality: mostly hybrid[1]. E.g. SOCs use **Suricata/Zeek** for known alerts + **ML on flows** to flag suspicious HTTPS C&C behavior.

Ref: [1] SANS Detection & Response Survey (2024).

ML-based NIDS as a prediction task



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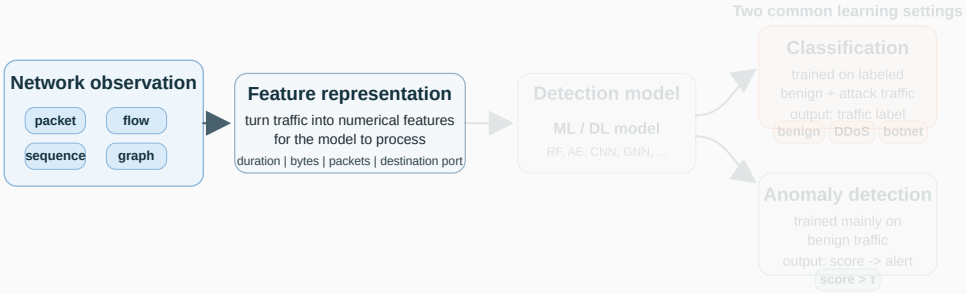

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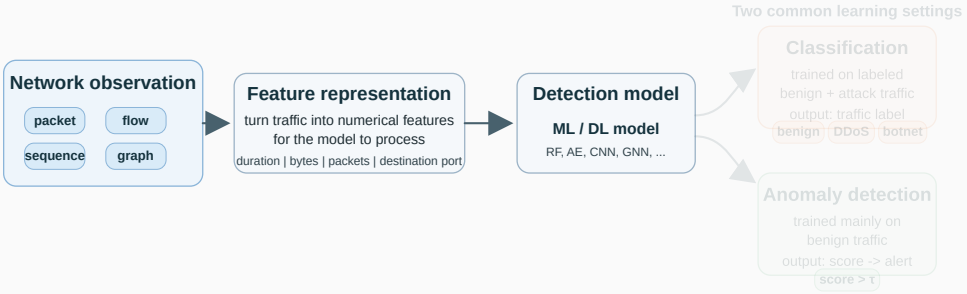

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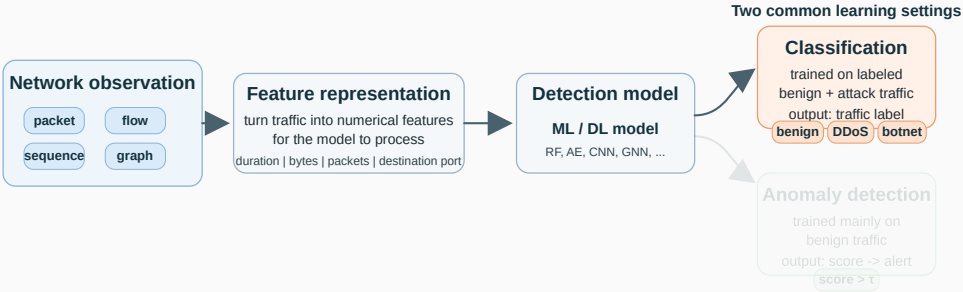

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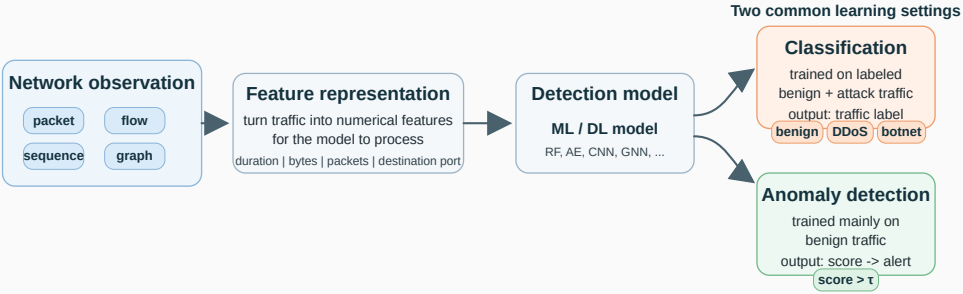

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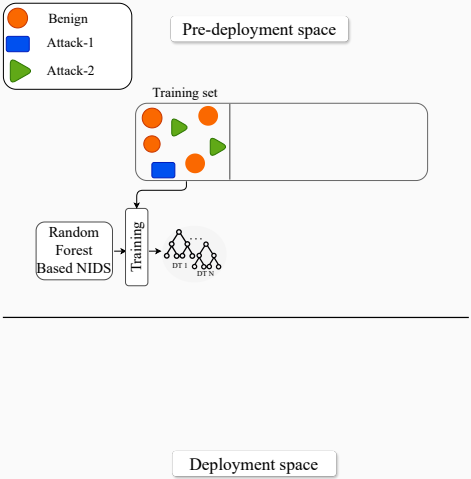

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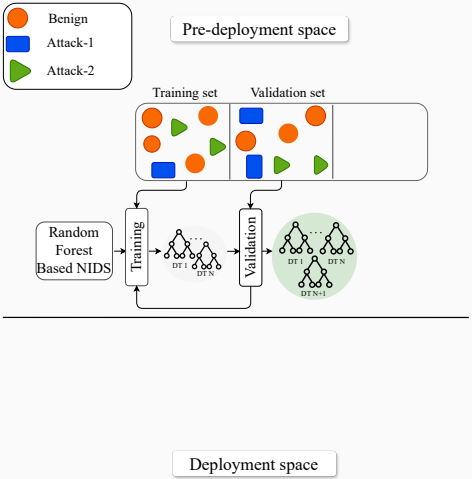

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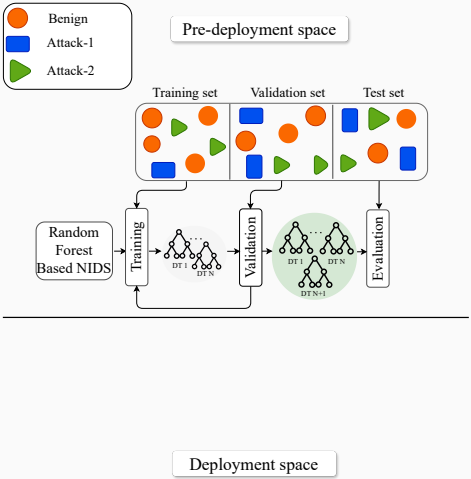

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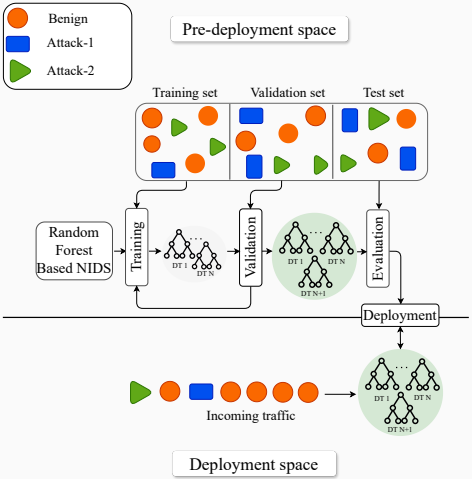
- The selected model is deployed to monitor live traffic.



How ML has been used to improve NIDS

- **A** Richer traffic representations ⟶ Flow-MAE[1], NetVigil[2]
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- **B** Better use of limited labels ⟶ Active-SVDD[3], INSOMNIA[4]
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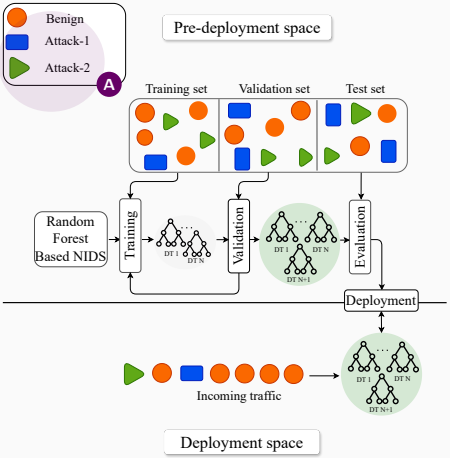

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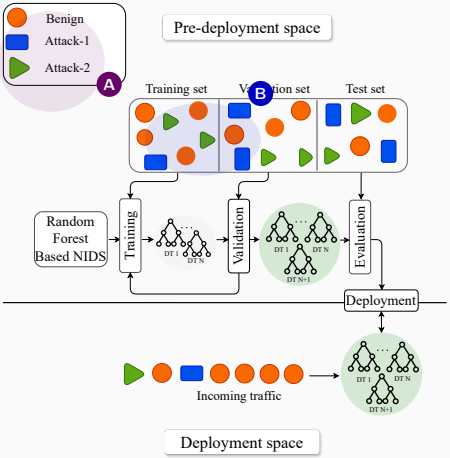

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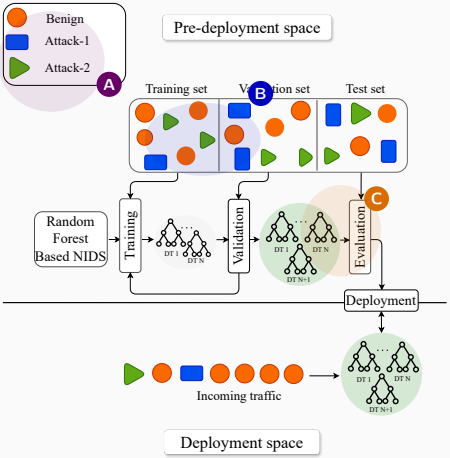

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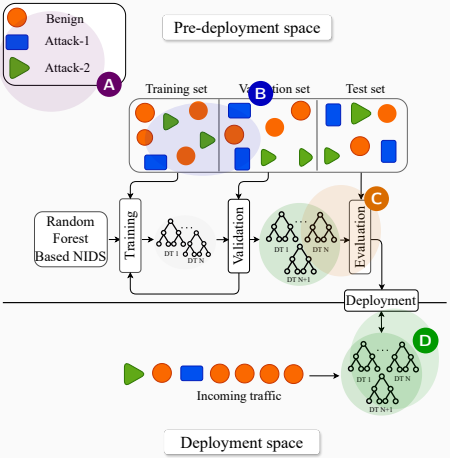

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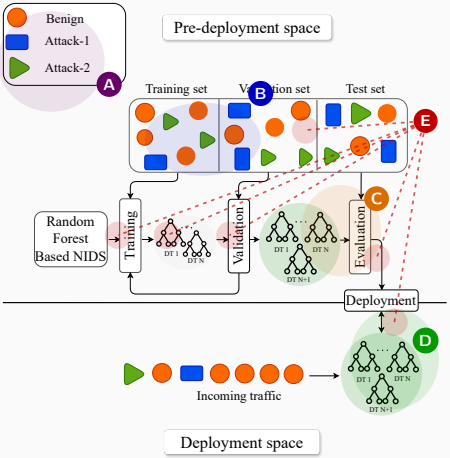

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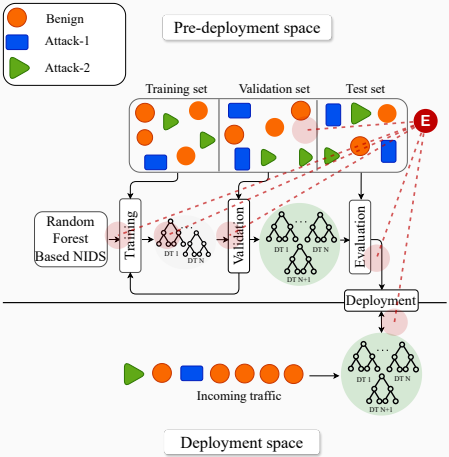

⇒ **The other axes optimize the detector inside a given setup, robustness questions the validity of that setup itself.**

Refs: [1] CADE [Yang et al., 2021, USENIX Security]; [2] OWAD [Han et al., 2023, NDSS].

Focus of this thesis: robustness in ML-based NIDS

Robustness in this thesis

- One fixed benchmark split is not enough.
- Robustness in operation: effectiveness under netwrok condition changes.
- Robustness in assessment: evaluation that truly stresses the detector.

$\Rightarrow$ **Robustness = adaptation + rigorous evaluation.**

Roadmap of the presentation

Part I Adaptation	Part II Evaluation
• Context change handling • Fast NIDS reconfiguration • Meta-learning-based auto-configuration	• Test-set difficulty assessment → relvealing NIDS weaknesses • Targeted augmentation of the test set when difficulty is low

- *Auto-tuning of Hyper-parameters for Detecting Network Intrusions via Meta-learning* (NOMS 2023, AnNet)
- *Automated Machine Learning Configuration to Learn Intrusion Detectors on Attack-Free Datasets* (LCN 2024, short paper)
- *TATA: Benchmark NIDS Test Sets Assessment and Targeted Augmentation* (ESORICS 2025)
- *An Empirical Study on Configuration Robustness of Unsupervised IDS* (NOMS 2026)

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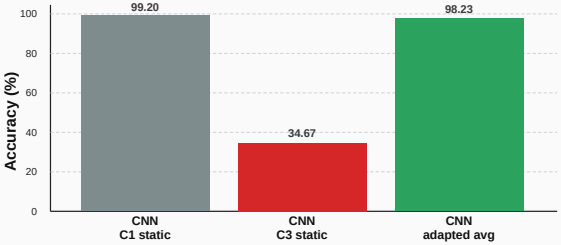
Why ML-based NIDS still need explicit adaptation

- Historical traffic for training; static detector at deployment.
- Assumes training and deployment traffic are statistically identical.
- Benign traffic evolves, attackers adapt → the network context changes.

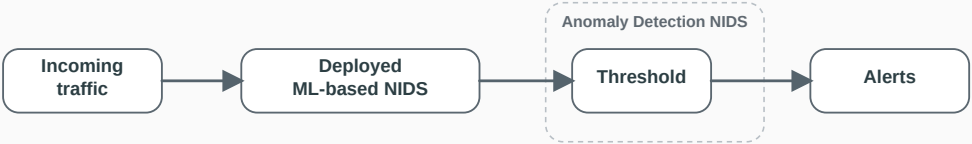


Source: Qiao et al., IEEE TII 2022, Table V/VI and Section IV-C

⇒ **Detection performance can degrade** ⇒ **explicit adaptation mechanisms are needed**

Adaptation = post-deployment update of (data | model | config | threshold) to maintain performance as the network context changes.

State of the art: main adaptation families



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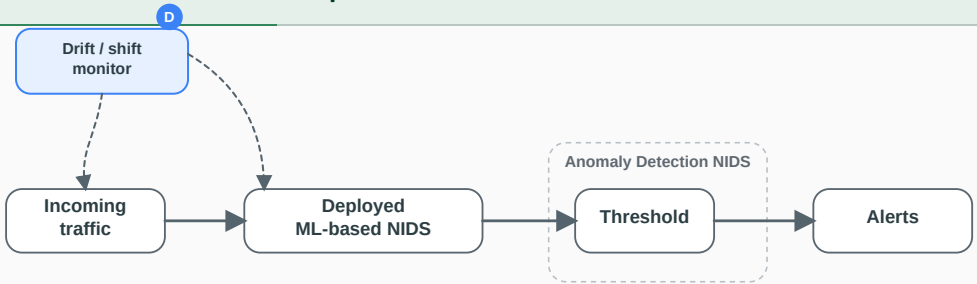

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monitor change in traffic or detector scores
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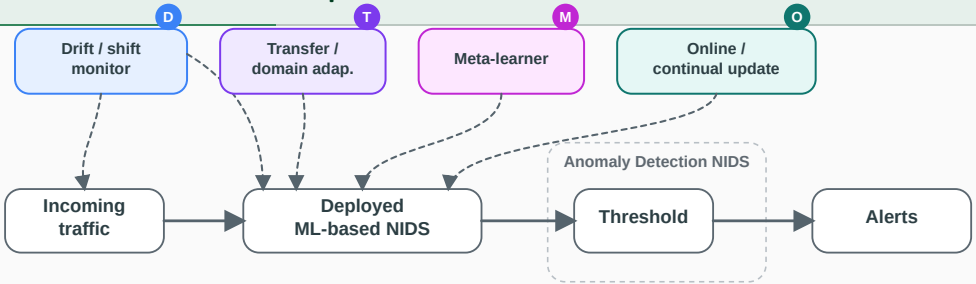

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reuse knowledge from another context
ADA, NAEF, DI-NIDS

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learn how to adapt quickly from prior tasks
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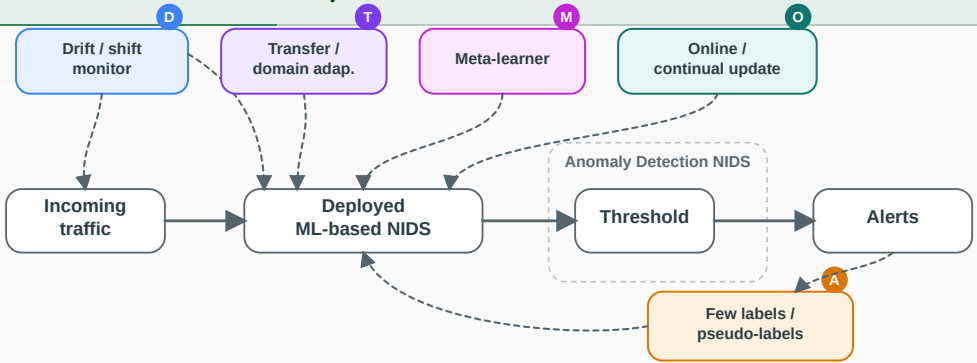

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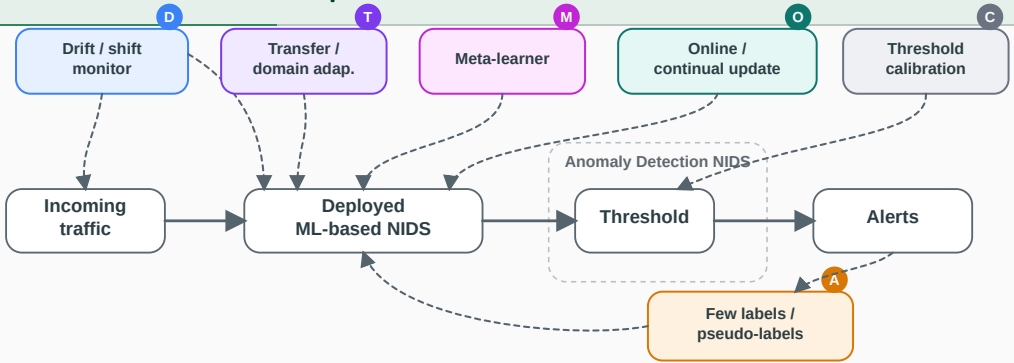

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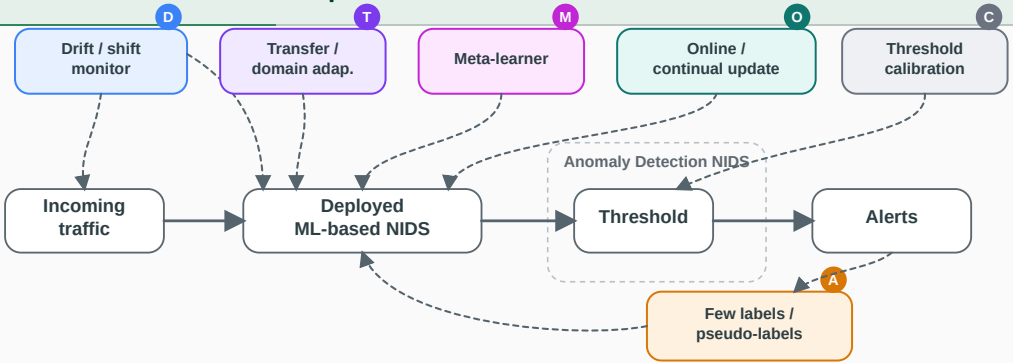

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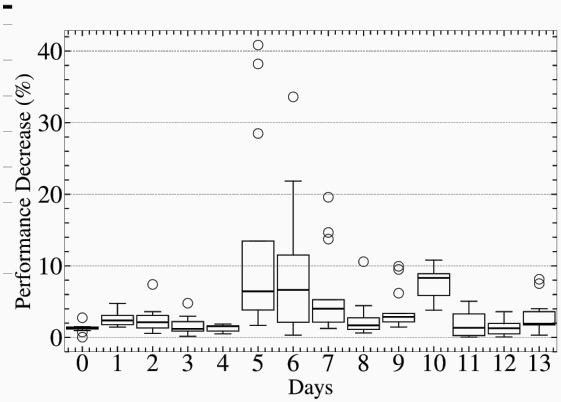
$\Rightarrow$ **Most methods adapt model, or threshold; configuration remains untargeted.**

Is hyperparameter configuration important?

- 14 daily IDS2017/IDS2018 datasets; each day a separate context.
- Can a day-t optimal config be reused on day t'?
- Each boxplot $\rightarrow$ performance drop from transferring another day's best config.
- Limited loss on some days; 20–40% degradation on others.



Source: Anser et al., IEEE/IFIP NOMS 2026, Figure 5

$\Longrightarrow$ **Configuration transfer helps in some contexts, but is not reliable in general.**

How is configuration handled today?

Methods

- Grid search, random search[1], Bayesian optimization (BO)[2].
- Repeated train-and-validate over candidate configurations.

Main drawback

- Offline and expensive.
- Restarted from scratch after each context change.

Real NIDS example[3]

- Bayesian optimization + binary Random Forest NIDS.

300	10 s	$\approx$ 50 min
iterations	per iteration	per daily context

$\Longrightarrow$ **Too slow for repeated post-deployment reconfiguration.**

Research question (RQ1)

Research question

How can we find a configuration of a ML-based NIDS in a timely manner while maintaining an acceptable level of detection performance?

From our analysis of hyperparameter impact

- A NIDS configuration method has to be applied at each network context change.
- A new network context is defined by a new attack in the network or by a topology change.

Source: Anser et al., *An Empirical Study on Configuration Robustness of Unsupervised IDS*, IEEE/IFIP NOMS 2026.

$\Rightarrow$ **Auto-configuration via meta-learning**

Auto-configuration via Meta-learning

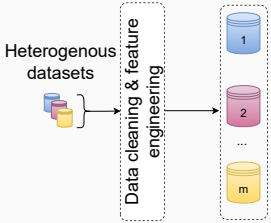
A. Meta-modeling

B. On the fly NIDS configuration

Trained NIDS

Auto-configuration via Meta-learning

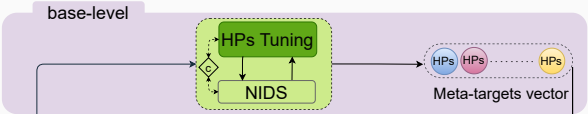
A. Meta-modeling

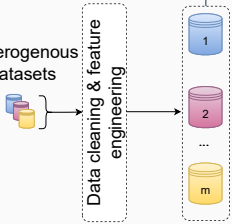


B. On the fly NIDS configuration

Auto-configuration via Meta-learning

For AD NIDS, tuning covers both HPs and threshold
HPs here = model HPs + threshold





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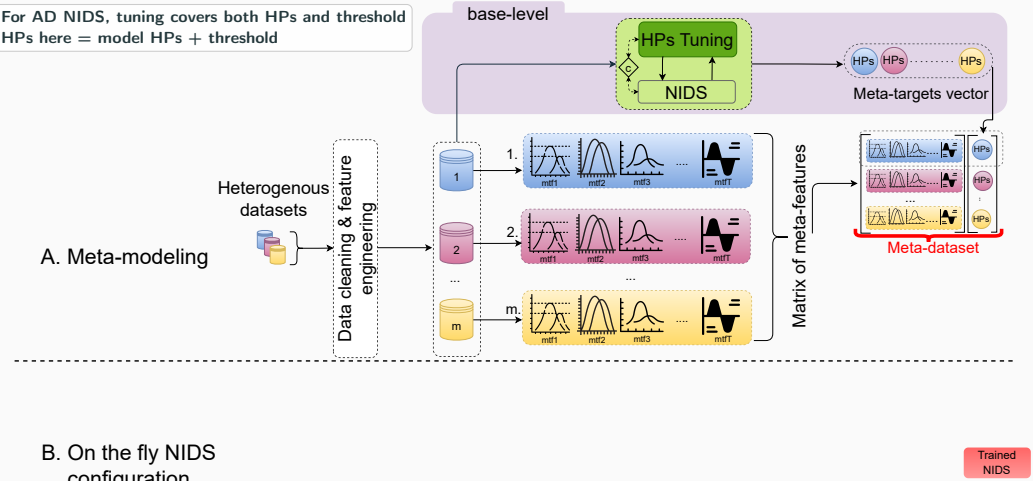

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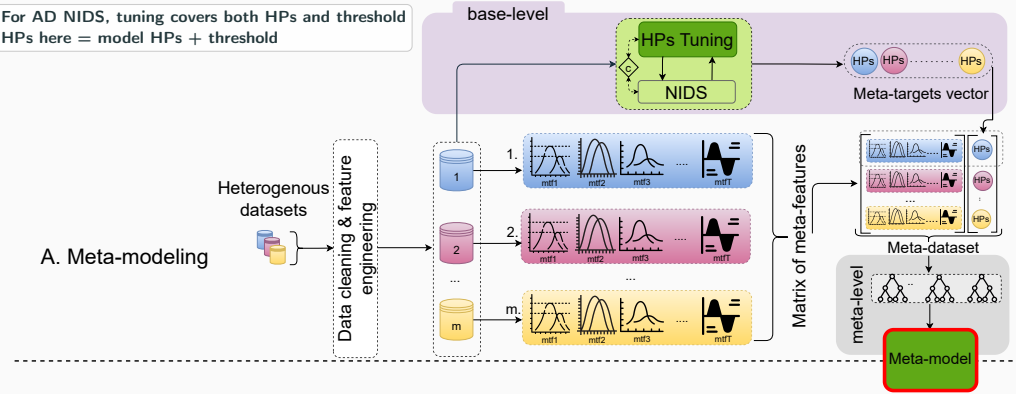

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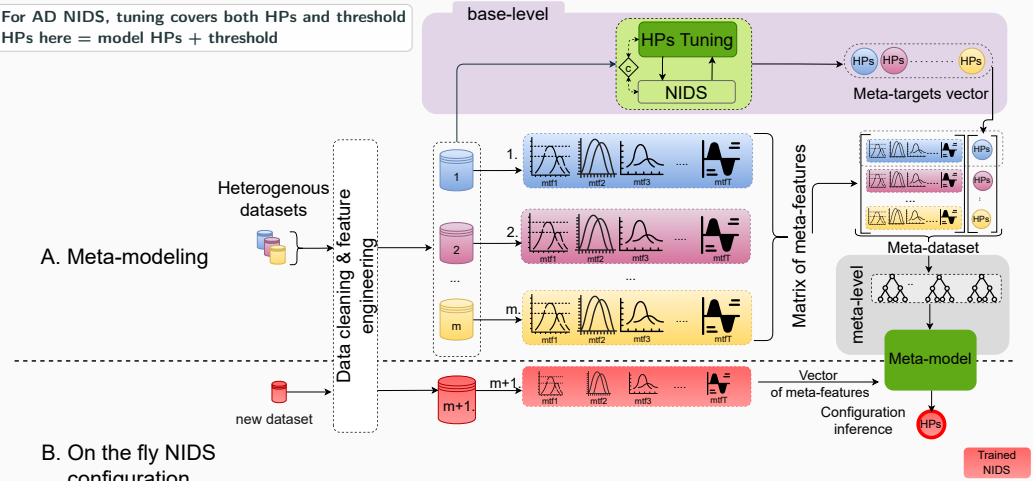

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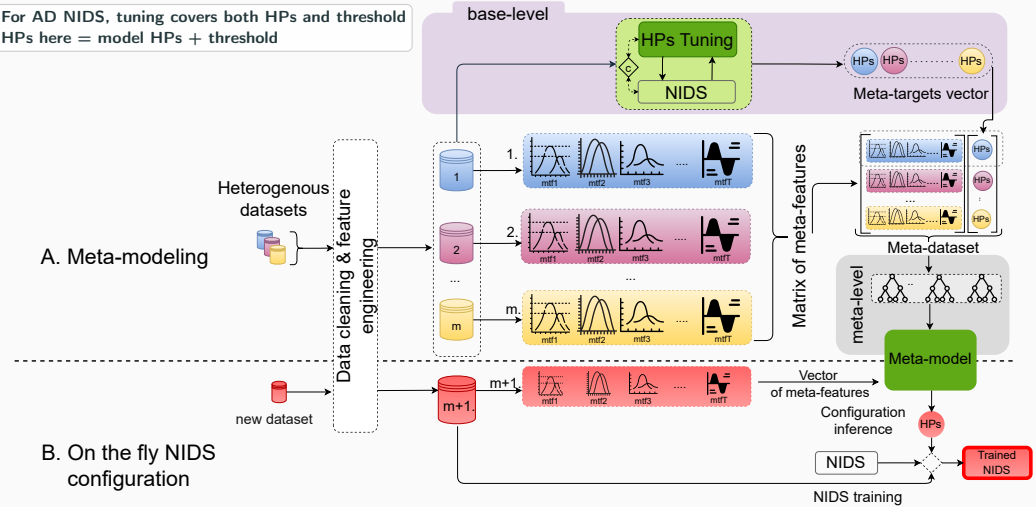


Methodology applied in two settings

Supervised setting

- Binary Random Forest on labeled benign/attack traffic.
- Contexts: **17 daily** IDS2017/IDS2018 datasets.
- Search space: **2 RF HPs** (‘max_features‘, ‘min_samples_leaf‘).
- BO baseline: **300 iterations/day**.

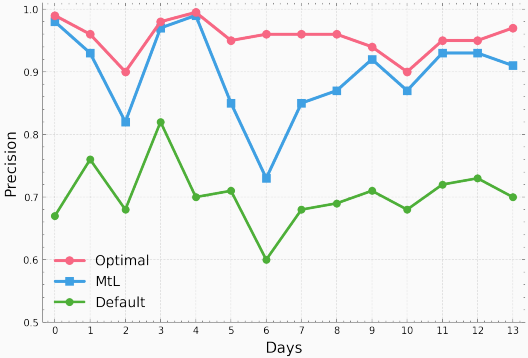
Unsupervised setting

- Isolation Forest-based AD IDS trained on attack-free data.
- Contexts: **14 daily** IDS2017/IDS2018 datasets.
- Search space: **7 iForest HPs + threshold**.
- BO baseline: offline daily optimum with labels.

$\Rightarrow$ **Same meta-learning pipeline, but different detectors, search spaces, and targets.**

Results: precision

- High overall precision in attack-free AD.
- Default configuration is much worse than MtL.
- Higher variability on days **2** and **5–8**.
- Harder contexts → weaker transfer.

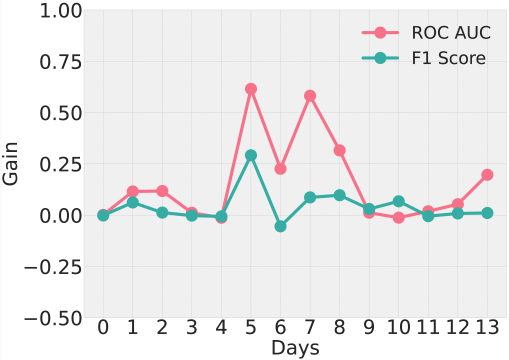


Unsupervised setting

⇒ **High precision overall; MtL stays much higher than default and closer to optimal on most days.**

Results: gain

- BO stopped at the same time budget as meta-learning.
- Gain = relative improvement: (meta-learning − baseline)/baseline.
- Strongest gains on harder days, especially **5–8**.
- Limited negative gains at equal cost.
- Supervised setting: large gains over the default.

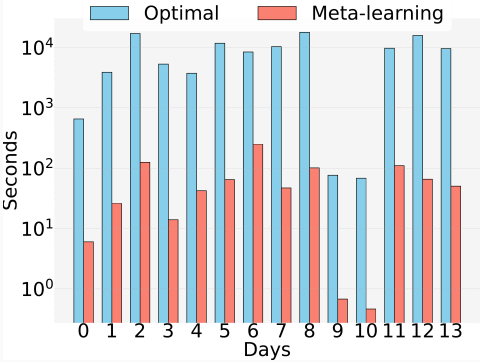


Unsupervised setting

⇒ **Cheaper than BO, yet competitive or better at equal budget.**

Results: configuration time

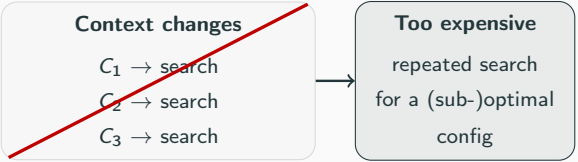
- Attack-free AD: seconds-to-minutes reconfiguration.
- BO runtime can exceed **100,000 s**.
- About **160x** faster overall.
- Supervised setting: about **9x** faster than finding a (sub-)optimal configuration.



Unsupervised setting

⇒ **Time advantage becomes decisive for online reconfiguration.**

Part I conclusion



- Meta-learning preserves strong detection performance.
- The operational benefit is time: visible in supervised NIDS and decisive in attack-free AD.

$\Rightarrow$ **Detailed analysis of the impact of configuration: Anser et al., *An Empirical Study on Configuration Robustness of Unsupervised IDS*, IEEE/IFIP NOMS 2026.**

Roadmap of the presentation

Part I Adaptation

- Context change handling
- Fast NIDS reconfiguration
- Meta-learning-based auto-configuration

Part II Evaluation

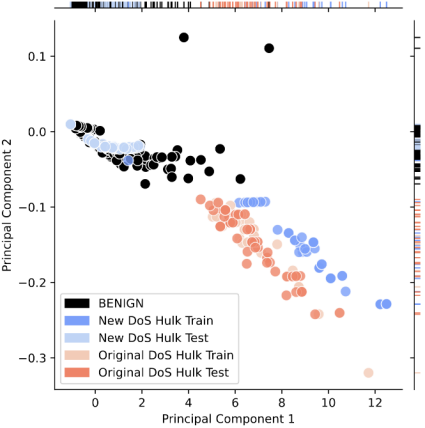
- Test-set difficulty assessment → relvealing NIDS weaknesses
- Targeted augmentation of the test set when difficulty is low

Auto-tuning of Hyper-parameters for Detecting Network Intrusions via Meta-learning (NOMS 2023, AnNet); *Automated Machine Learning Configuration to Learn Intrusion Detectors on Attack-Free Datasets* (LCN 2024, short paper).

TATA: Benchmark NIDS Test Sets Assessment and Targeted Augmentation (ESORICS 2025); *An Empirical Study on Configuration Robustness of Unsupervised IDS* (NOMS 2026).

Benchmark NIDS datasets — A Closer Look

- A single source dataset partitioned into $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$.
- Reported evaluation accuracies > **90%** in published works on public benchmarks (IDS2017, UNSW-NB15...) → solved problem or benchmark-induced illusions?
- A near-complete overlap of DoS-Hulk flows in IDS2017's $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$[1].

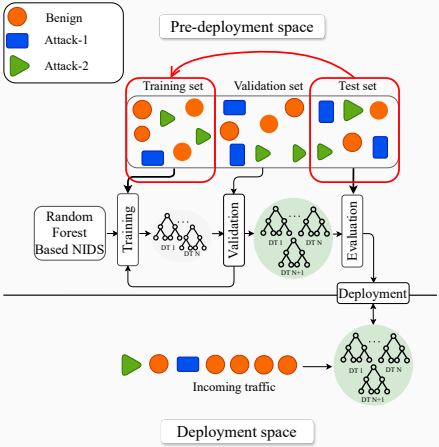


Overlap between $\mathcal{D}_{\text{test}}$ and $\mathcal{D}_{\text{train}}$ does not prevent overfitting → it can hide it.

[1] Flood et al., Bad Design Smells in Benchmark NIDS Datasets, IEEE EuroS&P 2024.

Focusing on $\mathcal{D}_{\text{test}}$ and its relation to $\mathcal{D}_{\text{train}}$

- Robust evaluation needs hard and corner cases.
- Pre-deployment evaluation relies on $\mathcal{D}_{\text{test}}$.
- Question: does $\mathcal{D}_{\text{test}}$ really stress the NIDS?
- $\mathcal{D}_{\text{test}}$ must be assessed relative to $\mathcal{D}_{\text{train}}$.



$\Rightarrow$ **This part focuses on $\mathcal{D}_{\text{test}}$ and its relation to $\mathcal{D}_{\text{train}}$.**

RQ2 — Targeted Augmentation

Given a **too-easy** initial $\mathcal{D}_{\text{test}}$, can we apply systematic augmentation to generate new, **realistic**, and **challenging** traffic that is specifically designed to expose the hidden weaknesses of the NIDS?

Research Questions

RQ1 — Test-set Assessment

How can we create a numerical score or a set of metrics to objectively measure the difficulty of a $\mathcal{D}_{\text{test}}$ for a NIDS?

RQ2 — Targeted Augmentation

Given a **too-easy** initial $\mathcal{D}_{\text{test}}$, can we apply systematic augmentation to generate new, **realistic**, and **challenging** traffic that is specifically designed to expose the hidden weaknesses of the NIDS?

Related Work

Method category	RQ1	RQ2	Key limitations
Neuron coverage			

Neuron coverage: Fraction of neurons whose activation exceeds a **preset threshold** at least once across the test set; **higher** $\rightarrow$ explore more of the model's **internal** logic.

Related Work

Method category	RQ1	RQ2	Key limitations
Neuron coverage			
Surprise coverage			

Surprise coverage: Measures how broadly the test set **differs from** what the model saw in training; **higher** → the test set includes **highly surprising inputs**.

Related Work

Method category	RQ1	RQ2	Key limitations
Neuron coverage			
Surprise coverage			
Mutation testing			

Mutation testing: Mutates the model and/or its training setup (e.g., weights, architecture, or hyperparameters), then checks if the test set can **distinguish the mutant from the original model**.

Related Work

Method category	RQ1	RQ2	Key limitations
Neuron coverage	✓	✗	Model access + NN-only + Ignores train–test alignment
Surprise coverage	✓	✗	+ + Proximity-only focus
Mutation testing	✓	✓	+ + Model-change sensitive Retrain per dataset + Synthetic data
Manual (Flood et al., 2024)	✗	✓	Unguided augmentation

G1 Model-related gaps **G2** Shortcomings in assessing $\mathcal{D}_{\text{test}}$ → RQ1 **G3** Weaknesses in $\mathcal{D}_{\text{test}}$ augmentation workflows → RQ2

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TATA (Test-set Assessment & Targeted Augmentation)

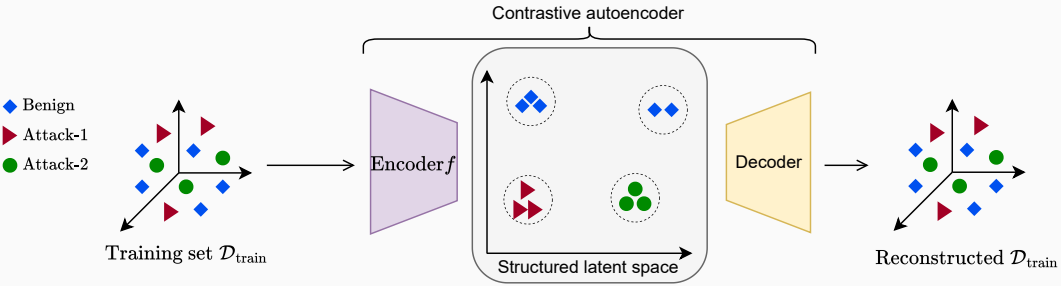
Goal: Close the gaps identified in related work and answer the research questions.

1. **Preliminary Phase** G1 : + +
   Approximate $\mathcal{D}_{\text{train}}$'s decision boundary while being model agnostic

2. **Phase 1 – Assesses $\mathcal{D}_{\text{test}}$'s quality (RQ1)** G2 : +
   Define three metrics (**Proximity**, **Diversity**, and **Scarcity**) to assess the **alignment** of $\mathcal{D}_{\text{test}}$ with respect to $\mathcal{D}_{\text{train}}$'s distribution.

3. **Phase 2 – $\mathcal{D}_{\text{test}}$ Augmentation (RQ2)** G3 : + +
   Reinforcement learning (RL) guides a testbed to generate **realistic traffic** that fills uncovered metric gaps, producing **a more challenging** $\mathcal{D}_{\text{test}}$.

Preliminary Phase: Approximating $\mathcal{D}_{\text{train}}$'s decision boundary



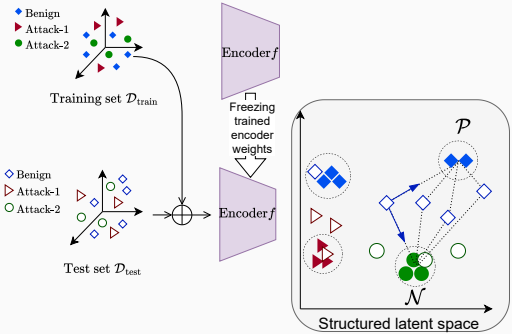
- Decision boundaries $\approx$ cluster borders in the latent space; these are captured in the encoder's weights.

Phase 1: Test-set Assessment

- Encode $\mathcal{D}_{\text{train}}$ & $\mathcal{D}_{\text{test}} \rightarrow \mathcal{Z}_{\text{test}}$ is located relative to $\mathcal{Z}_{\text{train}} \rightarrow$ we now work entirely in latent space $(\mathcal{Z}_{\text{train}}, \mathcal{Z}_{\text{test}})$.

- Proximity, diversity, and scarcity to characterize where $\mathcal{Z}_{\text{test}}$ lies in latent space relative to $\mathcal{Z}_{\text{train}}$ clusters.

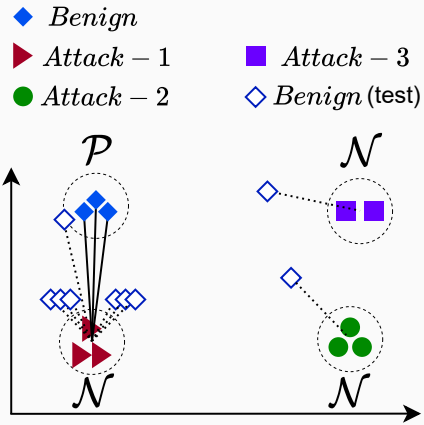
- For each test embedding z, we identify:
  - $P(z)$: the nearest **same-label cluster** (*positive cluster*)
  - $N(z)$: the nearest **different-label cluster** (*negative cluster*)



Proximity $P \in [0, 1]$, higher = nearer to different-label boundaries

- Quantifies $\mathcal{Z}_{\text{test}}$ closeness to $N(z)$; high proximity $\Rightarrow$ closer to a different label decision boundary $\Rightarrow$ harder cases.

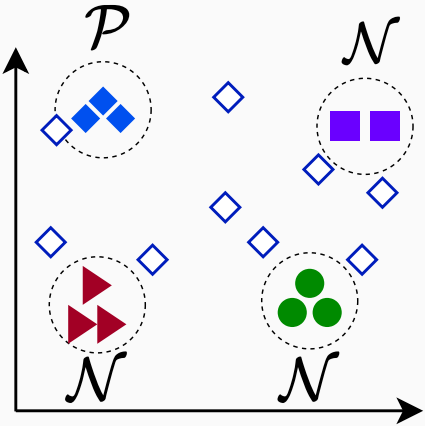
- P = **max** one-sided Kolmogorov–Smirnov statistic (over clusters): compares $\mathcal{Z}_{\text{test}}$ vs. $\mathcal{Z}_{\text{train}}$ distances to $N(z)$



Diversity $D \in [0, 1]$, higher = broader, less redundant coverage

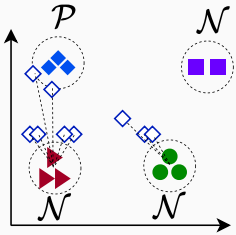
- Assesses the variety of $\mathcal{Z}_{\text{test}}$; higher diversity $\Rightarrow$ minimal redundancy $\Rightarrow$ broader coverage of the latent space.

- D = **Average** of normalized **Vendi Scores** (over clusters)

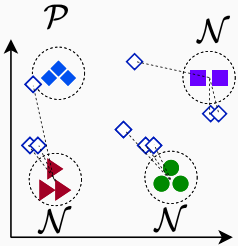


Scarcity $S \in [0, 1]$ higher = boundaries challenged more evenly

- Evaluates how uniformly $\mathcal{Z}_{\text{test}}$ are distributed among all potential negative clusters (i.e., decision boundaries).
- High scarcity $\Rightarrow$ a uniform spread $\Rightarrow$ all boundaries are exercised $\Rightarrow$ a more robust test set.
- S = **Average** of the **(1 - Gini coefficient)** (over clusters)

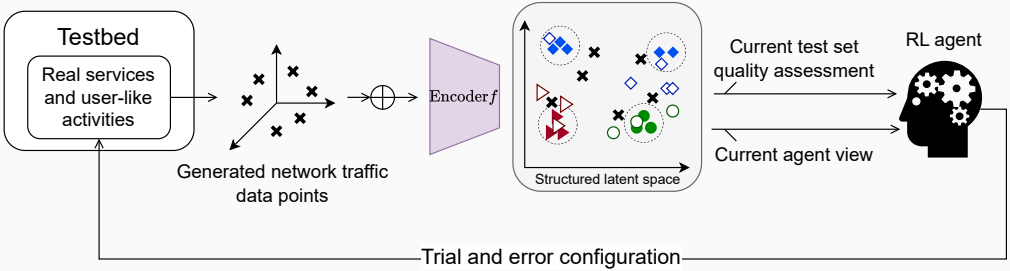


Bad scarcity



Optimal scarcity

Phase 2: Targeted Augmentation



- Start from Preliminary & Phase 1 outputs
- Configurable live testbed for realistic traffic generation.
- Legitimate multi-protocol traffic (HTTP/HTTPS, SSH, SMTP, DNS, . . .) and controlled attack traffic.

- **Training workflow**:

$$\boxed{\text{observe latent-space}} \rightarrow \text{select a configuration} \rightarrow \text{traffic generation} \xrightarrow{\text{embed}} \text{update } \mathcal{Z}_{\text{test}}$$

$$\rightarrow \text{compute proximity, diversity, and scarcity} \rightarrow \text{update policy } \pi \circlearrowleft \Rightarrow \pi^*$$

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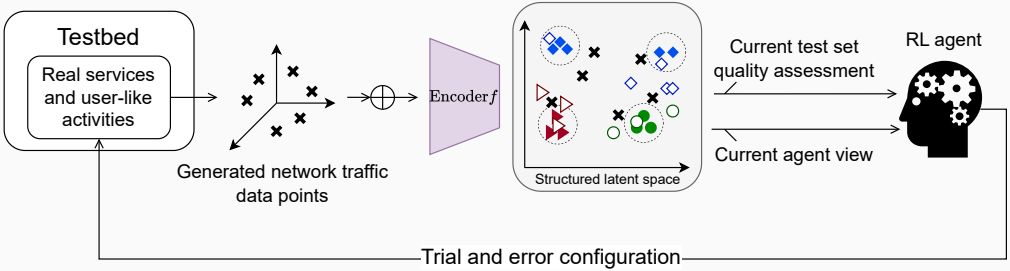

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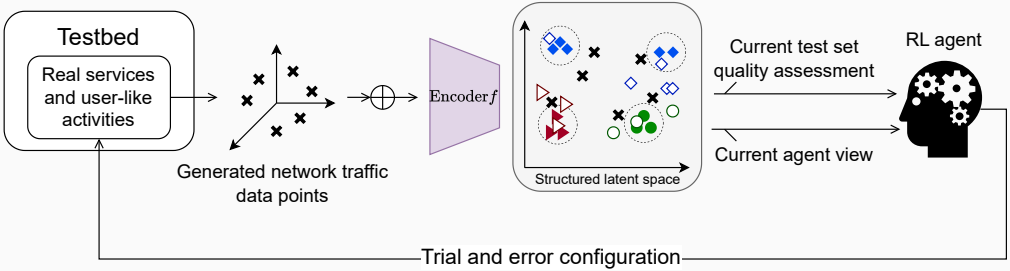

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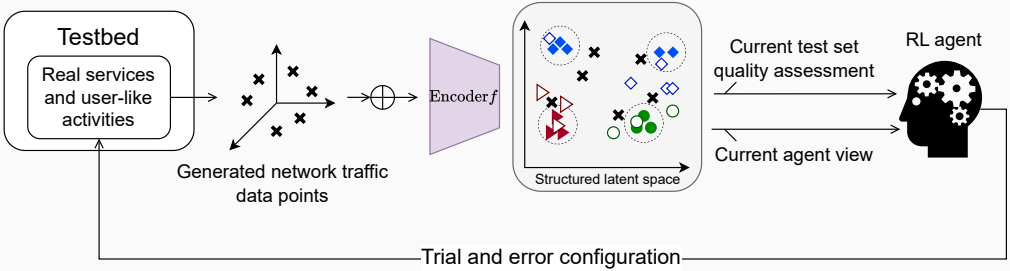

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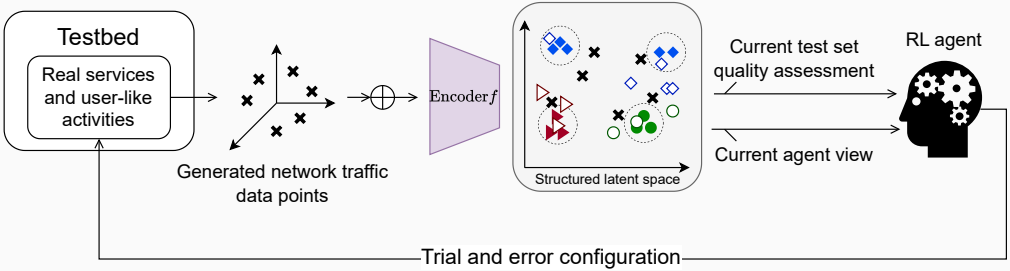

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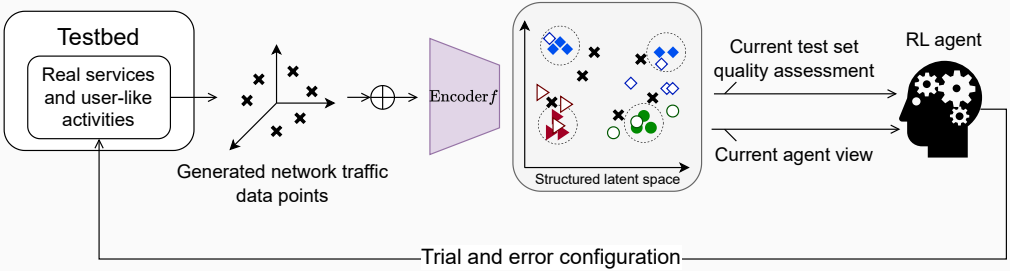

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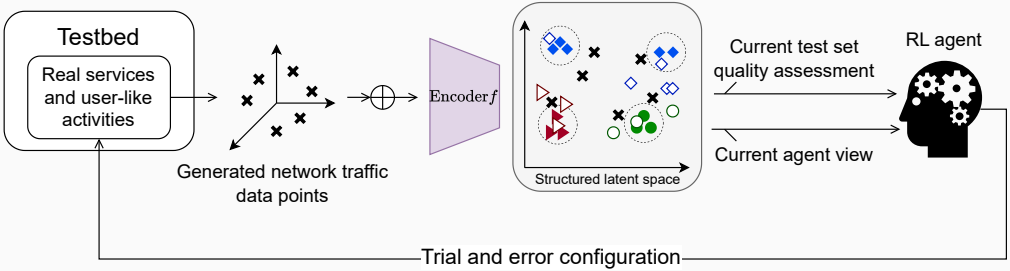

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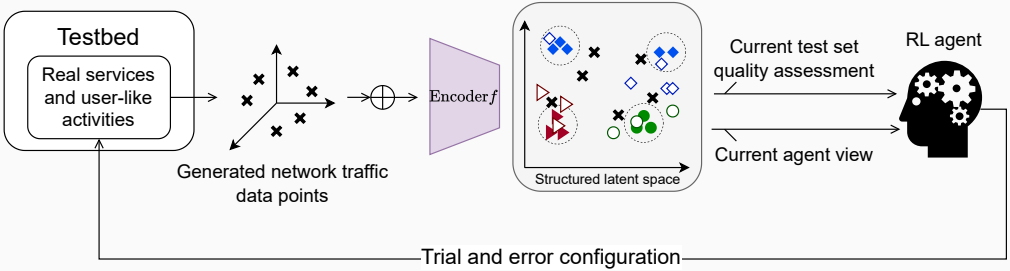
$$\rightarrow \text{compute proximity, diversity, and scarcity} \rightarrow \text{update policy } \pi \; \boxed{\circlearrowright} \; \Rightarrow \pi^*$$

Phase 2: Targeted Augmentation



- Start from Preliminary & Phase 1 outputs
- Configurable live testbed for realistic traffic generation.
- Legitimate multi-protocol traffic (HTTP/HTTPS, SSH, SMTP, DNS, . . .) and controlled attack traffic.

- **Training workflow**:

$$\text{observe latent-space} \rightarrow \text{select a configuration} \rightarrow \text{traffic generation} \xrightarrow{\text{embed}} \text{update } \mathcal{Z}_{\text{test}}$$
$$\rightarrow \text{compute proximity, diversity, and scarcity} \rightarrow \text{update policy } \pi \circlearrowleft \Rightarrow \boxed{\pi^*}$$

Experimental Setup

Datasets (60 / 20 / 20 split)

(Using the revised and corrected versions)

- **CIC-IDS 2017**: 0.93 M benign / 0.25 M attack flows
- **CSE-CIC-IDS 2018**: 59.8 M benign / 4.1 M attack flows

NIDS Models

- RF, SVM, 3-layer DNN (128-128-128 ReLU + Softmax)

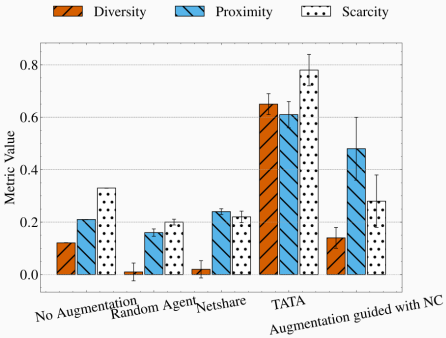
Agent Training

- Augmentation targets Benign SSH flows only

- Action space (client-side via `tc`): **Loss** $\in [5, 10]\%$ | **Jitter** $\in [4, 10]\,\mathrm{ms}$ | **Delay** $\in [10, 40]\,\mathrm{ms}$ | **Duplication** $\in [0.1, 5]\%$ | **Corruption** $\in [0.1, 10]\%$ | **Reordering** $\in [0.1, 50]\%$ | **Correlation** $\in [50, 100]\%$.

Test-set Augmentation — Metrics

- **Protocol:** 20 eval episodes on IDS2017; no reward/updates; add Benign SSH flows; report mean±sd.
- **Baselines:** Random, NetShare (GAN), RL + Neuron Coverage (NC).



Test-set quality metrics (higher is more challenging/covered).

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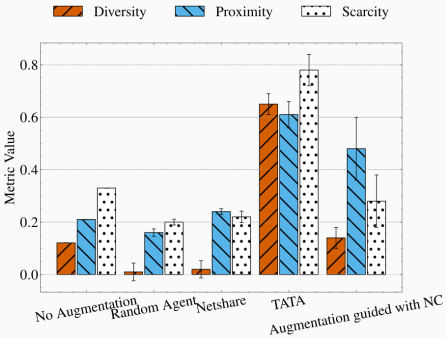
- **Quality metrics:**
  - **TATA:** $D + \sim 437\%$; $P + \sim 190\%$; $S + \sim 136\%$
  - **NC:** $D \approx 0$; $P + \sim 129\%$; $S \approx 0$
  - **Random/NetShare:** $D \downarrow$ to ≈ 0; $P \approx 0$; $S \downarrow$



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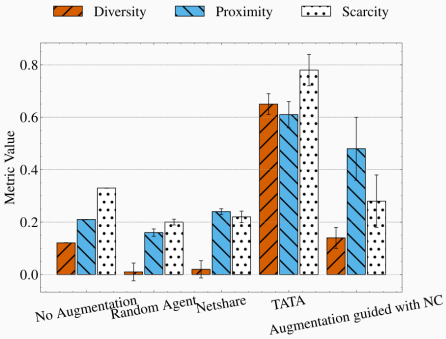
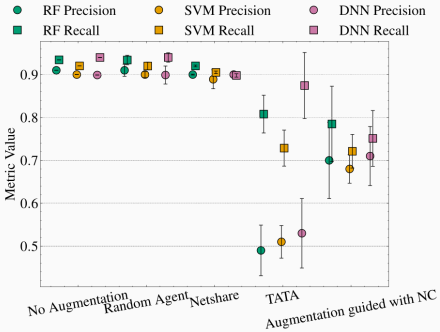
What is the impact on NIDS performance?



Test-set quality metrics (higher is more challenging/covered).

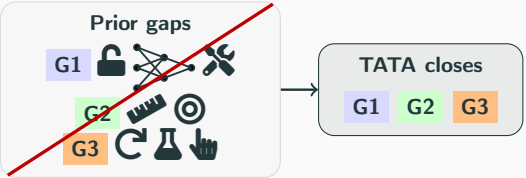
Test-set Augmentation — Impact on NIDS

- **NIDS impact (RF example):**
  - **TATA:** Precision $-\sim 46\%$; Recall $-\sim 13\%$
  - **NC:** Precision $-\sim 23\%$; Recall $-\sim 16\%$
- **Note:** higher scarcity spreads errors across boundaries $\Rightarrow$ FP $\uparrow \Rightarrow$ precision $\downarrow$.
- $\Rightarrow$ **proximity is the key driver;** scarcity and diversity are complementary (spread errors, avoid redundancy)

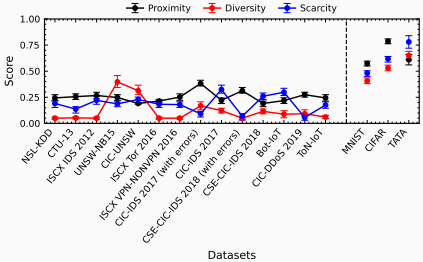


Impact on detection metrics (example: RF).

Part II conclusion



- Across public NIDS benchmarks, metrics remain mostly below **0.5** ⇒ limited test-set difficulty.



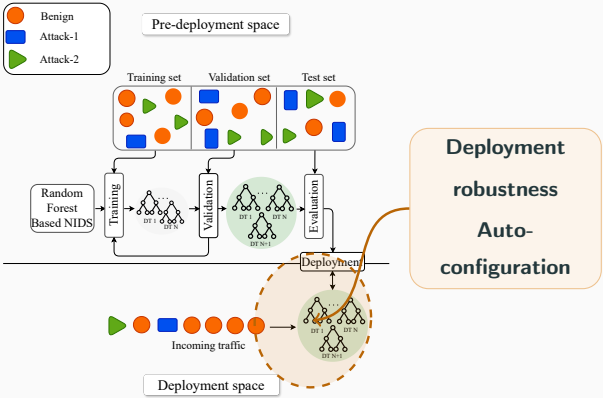
Public NIDS benchmarks ordered on the x-axis by release year (2009–2020)

Conclusion and perspectives

General conclusion

Auto-configuration perspectives

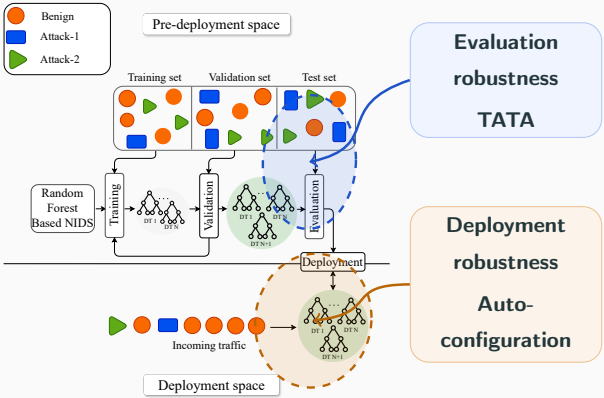
- Closed-loop reconfiguration under context change.
- Cold-start mitigation across organizations. → privacy-preserving and federated configuration learning.



General conclusion

TATA perspectives

- Reuse hard cases for detector hardening.
- Beyond single-protocol, flow-based settings.
- Larger and richer stress-test testbeds.



$\Rightarrow$ **Robustness in this thesis = rigorous evaluation + adaptive configuration.**

Domain Perspectives

General takeaway

- Robust ML-based NIDS require both stronger pre-deployment evaluation and adaptive post-deployment configuration.

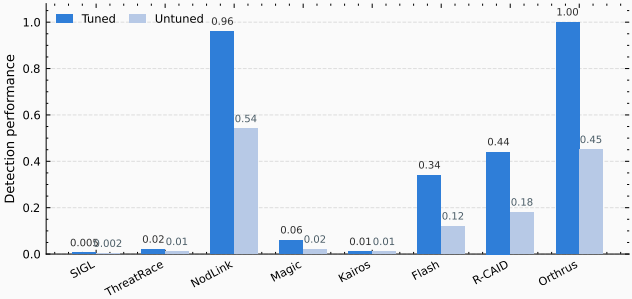
Perspectives

- Move from one-shot benchmarking to continuous evaluation and adaptation loops.
- Neuro-symbolic NIDS are a promising direction for robust and explainable detection.
- Interpretable rules can encode security invariants, support stress testing, and bound safer reconfiguration.

Questions?

Backup: provenance-based IDSs

- Bilot et al. compare 8 PIDSs.
- Tuned vs untuned, same task.
- Large gaps across several models.
- Mostly graph-based models.



Source: **Bilot et al., USENIX Security 2025, Figure 5**

$\Longrightarrow$ **Example on provenance-based IDSs: tuning matters across graph-based models too.**